

Can Residual Learning Improve GARCH?

Eddie Liu

A controlled test of the original additive LSTM, distinct from the log-ratio alternative.

Executive Summary

The original additive-residual LSTM, with its fixed output floor, has 5.48% lower observed MSE than corrected GARCH: 1.512933e-07 versus 1.600623e-07.

The 95% paired temporal interval for the MSE difference includes zero; the observed ordering is not a statistically resolved improvement in this sample.

The unbounded model produces 60 nonpositive forecasts. The primary reconstruction clips 60 forecasts to 1e-12; its QLIKE is 4.5780e+05, versus -8.46125 for GARCH. The MSE ranking alone does not establish valid or well-calibrated variance forecasts.

Same-date comparison

Forecast	MSE / 10 ⁻⁷	MSE change	QLIKE	Floor hits
GARCH-t	1.60062	+0.00%	-8.46125	0
Pure LSTM	1.93876	+21.13%	-6.44830	0
Feature LSTM	1.96415	+22.71%	-6.60657	0
Ratio LSTM	1.86316	+16.40%	-6.65949	0
Additive LSTM: raw	1.51343	-5.45%	Undefined	60
Additive LSTM: clipped	1.51293	-5.48%	4.5780e+05	60

Table 1. Lower MSE and QLIKE are better. MSE change is relative to GARCH; negative means improvement. The raw and clipped rows represent one model. Raw floor hits count values at or below 1e-12, not an applied raw-output floor. Undefined QLIKE is not replaced with a favorable score.

Source: additive-residual-v1 and unchanged corrected-v1 forecasts; 3,738 common dates, 2010-02-05 to 2024-12-11. One primary seed schedule; previously examined historical data.

01 / Restore the Original Question

The previous corrected run changed the additive residual model into a multiplicative log-ratio model to guarantee positive forecasts. That change did not test whether the original additive residual objective improves overall variance MSE. This follow-up supplies that missing control.

Target and reconstruction

$$\begin{aligned}e_t &= y_t - g_t \\h_t^{\text{raw}} &= g_t + \text{predicted } e_t \\h_t^{\text{clipped}} &= \max(h_t^{\text{raw}}, 10^{-12})\end{aligned}$$

Here y is squared daily log return and g is the corrected GARCH forecast, not a return residual or standardized GARCH innovation. Signed residuals receive training-only affine standardization; they are not logged and their sign is not constrained.

Control	Specification
Information and sample	178 expanding-window folds; 3,738 common test dates. Sequences end at $t-1$. Initial training has 756 sequences; validation has 252 observations; test blocks have 21.
Network and inputs	LSTM, eight units, 21-row lookback, dropout 0.1. The same lagged returns, squared/absolute returns, rolling volatility, and GARCH-feature transformation as the prior hybrid models.
Optimization	Linear output; standardized signed-residual MSE; Adam 0.001; batch 64; maximum 35 epochs; patience six; restored best-validation checkpoint.
Randomness and budget	Seed 42 plus split ID; deterministic CPU operations. Same epoch cap and stopping rule, not identical realized epoch counts. Mean epochs: 19.43.
Reused evidence	Corrected GARCH and all benchmark prediction files remain byte-identical. No test-selected shrinkage, alternative seed, or floor is introduced.

Why residual MSE is the relevant objective

$$(y - h^{\text{raw}})^2 = (e - \text{predicted } e)^2$$

The identity is verified numerically. Within each fold, affine target standardization only rescales the squared-loss objective by a fixed positive factor. It does not imply that a finite-sample neural fit will improve out-of-sample error. Clipping changes the reconstruction and is reported separately.

02 / Uncertainty and Stability

Additive minus GARCH	Difference	95% lower	95% upper
Raw MSE / 10^{-7}	-0.08719	-0.24674	0.00119
Clipped MSE / 10^{-7}	-0.08769	-0.24731	0.00034
Clipped QLIKE	4.5781e+05	2.3958e+05	7.5455e+05

Table 2. Negative differences favor the additive model. MSE differences are scaled by 10^{-7} ; QLIKE differences are unscaled. Paired stationary-bootstrap percentile intervals: average block length 21, 2,000 replications, seed 20260924.

The 95% paired temporal interval for the MSE difference includes zero; the observed ordering is not a statistically resolved improvement in this sample.

These are pointwise temporal intervals conditional on the trained forecasts. They do not include variation across training seeds, repeated research choices, or future regimes. Block lengths 5 and 63 are sensitivity checks retained in the aggregate evidence.

Calendar-period MSE

Calendar period	n	GARCH MSE	Raw MSE	Clipped MSE
2010-2016	1739	0.48819	0.45999	0.45972
2017-2019	754	0.27962	0.26222	0.26184
2020-2021	505	8.99661	8.45199	8.45007
2022-2024	740	0.51356	0.52878	0.52863

Table 3. All three MSE columns are divided by 10^{-7} . These calendar periods were retained from the previous plan. No dates are excluded from the headline comparison.

How concentrated is the observed difference?

The clipped model has lower MSE in **98 of 178 test blocks**. The ten largest absolute daily MSE differences account for **42.36%** of total absolute daily differences. Both are descriptive diagnostics, not independent hypothesis tests.

Those ten dates contribute $-3.757654e-09$ to the full-sample mean difference. Selecting or deleting influential dates after seeing the losses would answer a different question; the reported ranking uses every date.

03 / Validity and Research Judgment

The unbounded model produces 60 nonpositive forecasts. The primary reconstruction clips 60 forecasts to 1e-12; its QLIKE is 4.5780e+05, versus -8.46125 for GARCH. The MSE ranking alone does not establish valid or well-calibrated variance forecasts.

Output floor	MSE / 10 ⁻⁷	QLIKE	Mean h / mean y
1e-12	1.51293	4.5780e+05	1.051
1e-10	1.51293	4.5704e+03	1.051
1e-08	1.51293	37.32935	1.051
1e-06	1.51292	-7.96141	1.051

Table 4. Only 1e-12 is the primary, original-notebook floor. Other fixed floors are numerical sensitivity diagnostics, not candidate models selected by test performance. QLIKE uses $\log(h) + y/h$, with the established numerical floor for zero targets.

MSE improvement is not enough for deployment

The additive objective directly addresses the original question, unlike log-ratio MSE. However, an unconstrained correction may create invalid variance forecasts. A fixed floor can prevent a numerical domain error without resolving the statistical problem. Any claim must distinguish point-estimate MSE, uncertainty, positivity, and QLIKE.

Reproducibility

The public repository includes the pre-scoring plan, additive runner, evaluator, tests, PDF builder, and aggregate hashes. Source inputs and all prior benchmark forecasts are verified unchanged. Raw and clipped outputs are retained separately; the residual-MSE identity, checkpoint diagnostics, and complete dates are checked.

Run `scripts/run_additive_residual.py` against the existing `corrected-v1` directory, then `scripts/evaluate_additive_residual.py`. Build this report from `reports/additive_results.json` using `reports/build_additive_report.py`. Raw market data and complete predictions are local and not bundled; input hashes identify, but do not replace, those files.

Limits and provenance

One equity index, one seed schedule, a noisy squared-return proxy, and repeatedly examined historical data. No return-alpha strategy, portfolio P&L, or production validation is claimed. The previous nine-specification study follows this note in the combined PDF and remains a distinct experiment.

The underlying project originated in collaborative research. This revised portfolio report is presented by Eddie Liu; contributor names and course identifiers are omitted without implying sole original authorship.

github.com/SRG-eddieliu/rnn-volatility-lab